

Smart Street Light Feeder Panel + CMS

Project Report: design, wiring, parts list, build and testing

Rev B · 10 October 2026

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1. Project overview

ITEM	DETAIL
Project	CMS: Central Monitoring System for street lights (tender)
Scope	160 smart feeder panels, ~40,000 poles (~250 poles per panel)
Panel	3-phase, 10 kW, 4G/LTE, energy meter data on CMS dashboard
Prepared for	RAHUL
Repository	github.com/roxjaglan000-ui/cms (PR #1 panel, PR #2 CMS software)
Status	Design complete. Firmware not yet compiled or tested on hardware.

Ye report tender ke **3-phase, 10 kW Smart Feeder Control Panel** ka poora design hai. Panel street lights ko time ya sunset/sunrise ke hisaab se ON-OFF karta hai, 3-phase energy meter padhta hai, faults pakadta hai, aur 4G se CMS dashboard ko data bhejta hai. Dashboard se panel ko turant command bhi de sakte hain.

Is report se aap ye kaam kar sakte ho:

1. Parts kharidna (Section 6)
2. Panel banana aur wiring karna (Sections 3–5, 9)
3. Firmware daalna aur CMS jodna (Sections 7–8)
4. Har panel ki testing aur site commissioning (Section 10)
5. Tender bid ka hisaab (Section 11)

1.1 Tender requirements

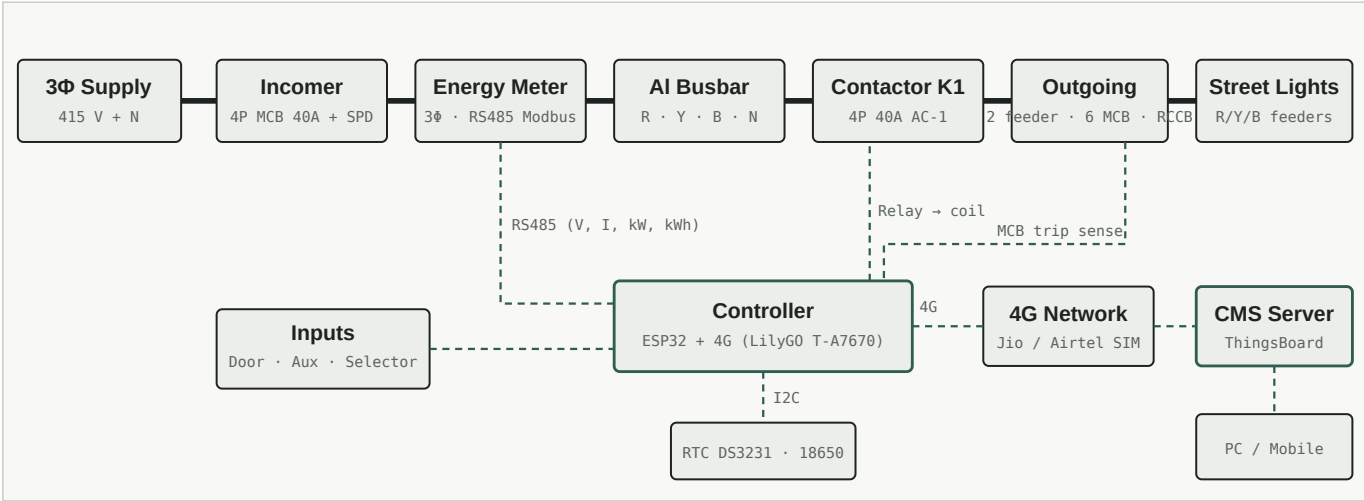
SITC of 3 Phase, 10KW Smart Feeder Control Panel for street light complete with suitable incomer & outgoing & sophisticated with following features & 5 years comprehensive warrantee.
a) Feeder Control panel shall communicate with application using 4G/LTE/5G network. There should not be any delay in response while using 4G/LTE/5G network.
b) Compartmentalised Aluminium Bus Bar with full neutral Feeder Panel with suitable incomer and outgoing of MCB shall have application support for:
- Scheduling of time of day, day of week/month, date
- Alarming and flexible reporting
- Data logging with historical data repositories
- Failure detection and reporting
- Event and fault Logging
c) The Panel shall have incorporating 3 phase energy meters who's data shall be received on the dashboard of CMS.
Supplying and fixing of hot dip galvanized octagonal poles of 12 mtrs. height flange mounted on the existing cement concrete pedestal i/c transportation charges, supplying and fixing of the following in the window box provided for cable termination. (in the case of accidental damaged)
Supply & fixing 6 mm thick laminated mounted sheet inside the window i/c providing mounting arrangement etc. complete as reqd.
Supply and fixing 63 amps. terminal block suitable for 16/25 sq. mm conductor- 4 Nos. on DIN channel i/c providing and fixing DIN channel of suitable length on the existing laminated sheet.
Supply & fixing of 6 amp. SP MCB 'C' characteristics Curve - 2 Nos, earth stud -1 Nos, at suitable height one nos. washers, springs washers & nuts.

Tender spec (screenshot)

#	REQUIREMENT (FROM SPEC)	COVERED IN
1	SITC of 3-phase, 10 kW smart feeder control panel for street lights, with incomer, outgoing and 5-year comprehensive warranty	Sections 2–6

#	REQUIREMENT (FROM SPEC)	COVERED IN
a	Panel communicates with the application over 4G/LTE/5G, with no delay in response	4G modem + persistent MQTT connection, commands applied immediately (Section 7)
b	Compartmentalised aluminium busbar with full neutral, MCB incomer and outgoing	Section 3
b1	Scheduling by time of day, day of week/month, date	ASTRO, SCHEDULE (weekday mask) and special-date modes (Section 7)
b2	Alarming and flexible reporting	ThingsBoard alarm rules and reports (Section 8)
b3	Data logging with historical data repositories	Telemetry every 5 min (configurable) plus instant events, offline buffer in flash, history in ThingsBoard
b4	Failure detection and reporting	Fault table in Section 7
b5	Event and fault logging	<code>event</code> telemetry key, RAISED/CLEARED per fault
c	3-phase energy meter data shown on the CMS dashboard	Selec MFM383A-C (or SDM630) over Modbus RS485
—	12 m hot-dip galvanised octagonal poles on existing pedestals	Civil/supply item, outside this design
—	6 mm laminated sheet inside the pole window box	Civil/supply item
—	63 A terminal block for 16/25 sq mm conductor, 4 nos, on DIN channel	Per pole, Section 6.2
—	6 A SP MCB C-curve, 2 nos, earth stud, per pole	Per pole, Section 6.2

2. System design



Block diagram

Kaise kaam karta hai: supply incomer, SPD aur energy meter se hoke aluminium busbar pe aati hai. Wahan se contactor K1 aur RCCB ke through do outgoing feeders tak jaati hai. Controller (ESP32 + 4G) har 5 second meter padhta hai, decide karta hai light ON ho ya OFF, relay se contactor chalata hai, MCB aur door jaise inputs check karta hai, aur MQTT se CMS ko data bhejta hai.

2.1 Load calculation (10 kW panel)

$I = P \div (\sqrt{3} \times V \times PF) = 10,000 \div (1.732 \times 415 \times 0.95) \approx \mathbf{14.6 \text{ A per phase}}$. LED drivers ka inrush zyada hota hai, isliye incomer aur contactor 40 A rakhe hain.

ITEM	VALUE
Supply	415 V, 3-phase + N
Poles per panel	$40,000 \div 160 = \mathbf{250}$ (≈ 83 per phase)
Max lamp jo 10 kW me fit hoga	$10,000 \text{ W} \div 250 = \approx \mathbf{40 \text{ W per pole}}$
Current per phase	$\approx 14.6 \text{ A}$

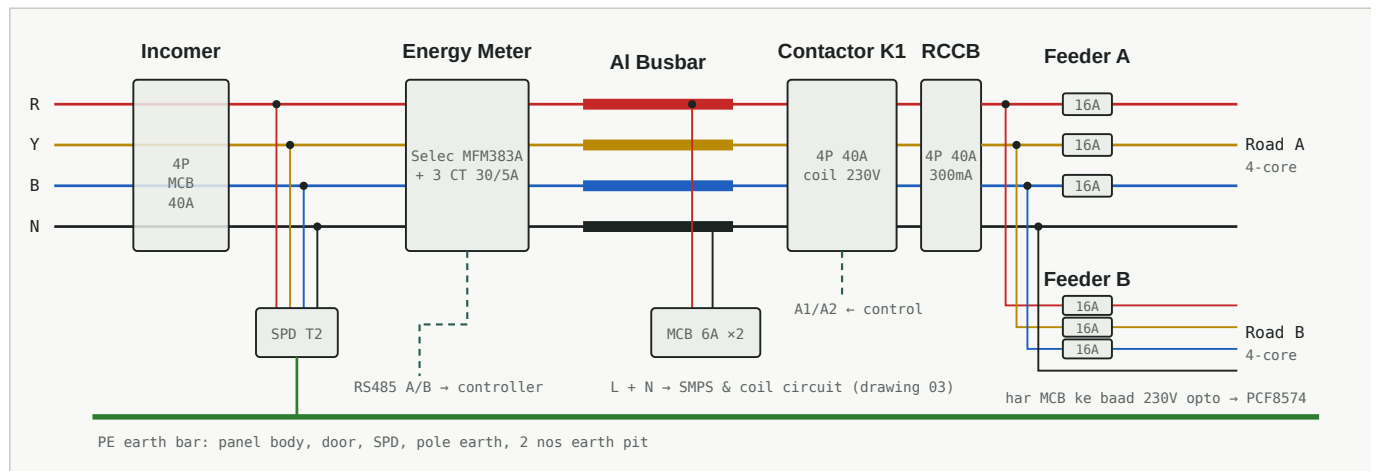
Sabse zaroori check: 10 kW panel me 250 pole tabhi chalenge jab har lamp lagbhag 40 W ka ho. Lamp 60–90 W ke hain to panel 15–22.5 kW ka banana padega (neeche table). Tender BOQ me lamp ki watt zaroor dekho.

2.2 Panel rating by lamp wattage (250 poles)

LAMP	LOAD	CURRENT / PHASE	INCOMER	RCCB 300 MA	CONTACTOR	CTS	FEEDER MCBS (×6)	IMAX	EXTRA COST VS 10 KW
40 W	10 kW	14.6 A	4P MCB 40 A	4P 40 A	4P 40 A AC-1	30/5	16 A	25	—
60 W	15 kW	22 A	4P MCB 63 A	4P 63 A	4P 63 A AC-1	50/5	20 A	35	₹3,000–4,500
90 W	22.5 kW	33 A	4P MCB 63 A (or 80 A MCCB)	4P 63 A	4P 63 A AC-1	50/5	25 A	50	₹4,000–6,000

Meter, controller, enclosure aur firmware same rehte hain. Sirf CMS se `setLimits` aur meter ka CT ratio setting badalna hai.

3. Power wiring



Power wiring (main circuit)

Raasta: Supply → 4P MCB 40 A incomer → SPD (earth tak) → Selec MFM383A-C energy meter (3 CT, R/Y/B pe) → aluminium busbar → contactor K1 → 4P RCCB 40 A 300 mA → 2 outgoing feeders (A aur B), har feeder me 3 SP MCB 16 A aur 4-core cable.

Control supply R-bus aur N se do 6 A MCB ke through li jaati hai (SMPS aur contactor coil ke liye).

Har 6 outgoing MCB ke baad ek 230 V opto channel hai. Ye controller ko batata hai kaun sa MCB trip hua. Agar 6 ke 6 ek saath band hon to RCCB trip hua hai.

Colour code (Indian): R lal, Y peela, B neela, Neutral kaala, Earth hara. Power wiring 6 sq mm Cu (terminal blocks pe spec ke hisaab se 16/25 sq mm), control wiring 1.5 sq mm Cu, signal wiring 0.5-0.75 sq mm.

3.1 Feeder design for 250 poles

- **2 three-phase feeder, 3 single-phase nahi.** Sadak ki har direction me ek 4-core cable, aur pole pe lamp R, Y, B, R, Y, B... baari baari jodo. Phase balance rehta hai aur voltage drop single-phase ke mukable lagbhag 6 guna kam.
- **RCCB 300 mA.** 2-3 km lambe feeder ke end pe phase-earth fault itna current nahi kheenchta ki 16 A MCB trip ho. RCCB earth fault kahin bhi ho, trip karega. CEA safety regulations bhi 5 kW se upar earth leakage protection maangte hain.
- **Har pole me 6 A MCB** (spec me hai) us pole ka lamp circuit bachata hai, taaki ek lamp ki kharabi sirf usi pole ko band kare.

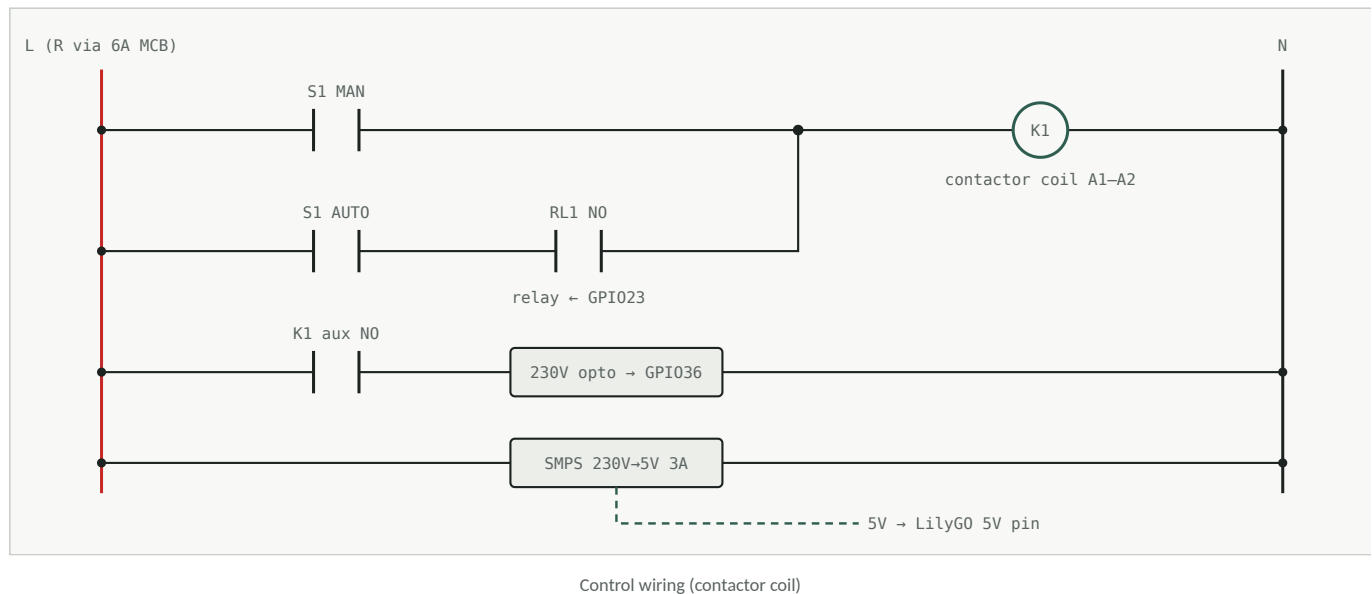
3.2 Feeder cable aur voltage drop

Example: har feeder pe 125 pole, 2.5 km sadak, R-Y-B balanced, aluminium armoured 4-core cable. Far end pe drop $\approx I \times R \times L \div 2$.

LAMP	CURRENT / PHASE / FEEDER	16 SQ MM AL	25 SQ MM AL	35 SQ MM AL	50 SQ MM AL
40 W	7.6 A	7.9%	5.0%	3.6%	—
60 W	11.4 A	—	7.4%	5.4%	4.0%
90 W	17.1 A	—	—	8.1%	6.0%

Lagbhag 5% tak rakho. Asli route length site survey se lekar dobara nikaalo. Panel apne area ke beech me rakho, taaki dono taraf ke feeder chhote rahein.

4. Control wiring

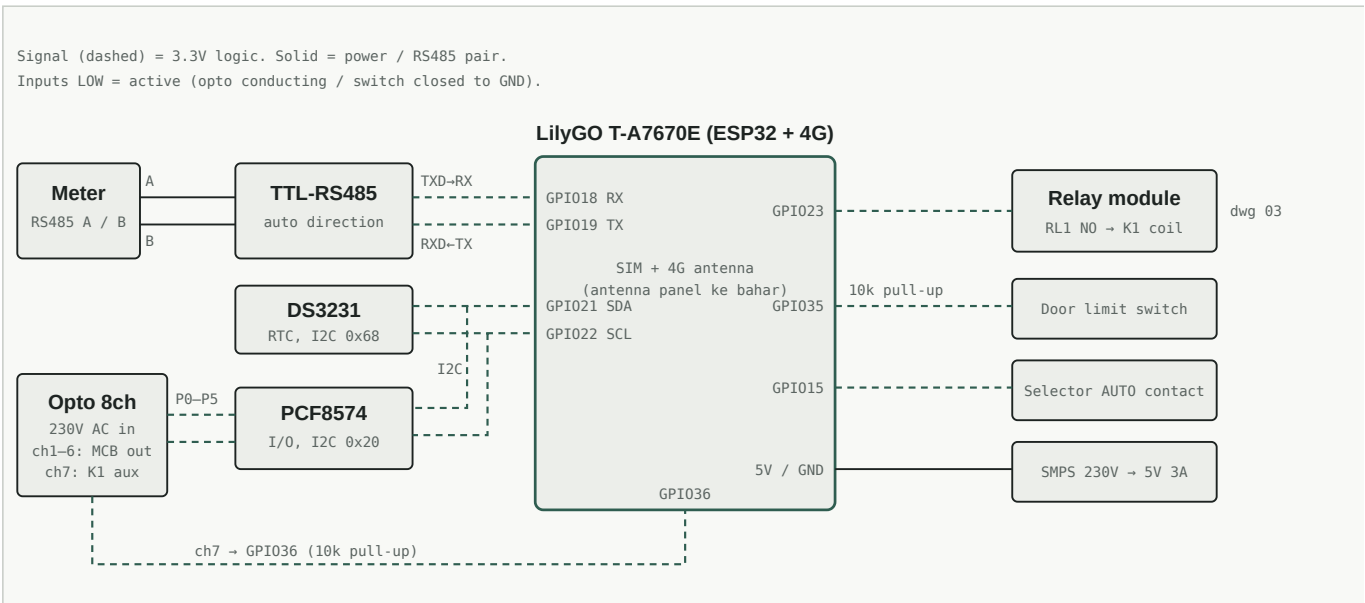


Selector switch ke 3 position hain:

- **AUTO:** contactor coil relay RL1 se chalti hai, jo controller (GPIO23) chalata hai.
- **OFF:** lights band.
- **MANUAL:** coil seedhe supply se. Controller kharab ho tab bhi lights jala sakte ho.

Contactor ka aux NO contact ek 230 V opto channel pe jaata hai, taaki controller ko pata chale contactor sach me band hua.

5. Controller wiring (low voltage)



Controller wiring

ESP32 PIN	CONNECTED TO	NOTE
GPIO18 / GPIO19	TTL-RS485 module TXD / RXD → meter A, B	Auto-direction module, twisted pair, 120 Ω at the end
GPIO21 / GPIO22	DS3231 RTC SDA / SCL	Keeps time without network
GPIO23	Relay module IN → RL1	Opto-isolated relay
GPIO35	Door limit switch	LOW = door closed, 10k pull-up
GPIO36	Contacto aux NO via 230 V opto	LOW = contactor on, 10k pull-up
GPIO21 / GPIO22 (I2C)	PCF8574 expander at 0x20, P0-P5 ← opto after MCBs A-R, A-Y, A-B, B-R, B-Y, B-B	LOW = supply present
GPIO15	Selector AUTO position (extra contact block)	LOW = AUTO
5V / GND	230 V → 5 V 3 A SMPS	The 4G modem draws up to 2 A peaks
SIM / antenna	Nano SIM, 4G antenna mounted outside the panel	Metal enclosures block signal

GPIO 34-39 sirf input hain aur inme internal pull-up nahi hota: GPIO35 aur GPIO36 pe 10 kΩ pull-up (3.3 V tak) lagao. Pin numbers LilyGO T-A7670E ke hisaab se hain; board aane pe uska pinout ek baar mila lo. 4G antenna panel ke bahar lagao, metal panel signal rokta hai.

6. Parts list (ek panel, 10 kW)

Rates October 2026 ke andaaze hain. Brand, shehar aur quantity ke hisaab se 15–25% upar-neeche ho sakte hain. Bid se pehle vendor quote zaroor lo.

#	ITEM	SPEC / EXAMPLE	QTY	APPROX ₹
1	Panel enclosure	CRCA 16/14 SWG, IP55, powder coated, ~500×400×200 mm, compartmentalised, from a local fabricator	1	5,000–6,500
2	Incomer	4P MCB 40 A C-curve (the spec asks for an MCB incomer)	1	1,600–2,200
3	Surge protection device	Type 2, 3P+N, 20 kA. Not in the spec; keep it to protect the 5-year warranty	1	1,500–2,000
4	Aluminium busbar set	R, Y, B, N strips + SMC insulators	1	1,200–1,800
5	3-phase energy meter	Selec MFM383A-C (RS485) + 3 CT 30/5 A. Indian make	1	3,800–4,400
6	Power contactor	4P 40 A AC-1, 230 V coil, + 1NO aux, reputed Indian brand	1	1,800–2,500
7	Outgoing MCB	SP 16 A C-curve, 3 per feeder × 2 feeders	6	900–1,200
7a	RCCB (earth leakage)	4P 40 A, 300 mA. Required by CEA safety regulations for loads above 5 kW; also protects the long feeders	1	2,000–2,800
8	Control MCB	SP 6 A C-curve	2	300
9	Selector switch	3-position AUTO-OFF-MAN, 22 mm	1	200–250
10	Indicator lamps	22 mm LED: R, Y, B, Light ON	4	240–300
11	Controller + 4G	LilyGO T-A7670E (ESP32 + A7670 LTE Cat-1) with antenna	1	2,300–2,800
12	Power supply	230 V AC → 5 V 3 A SMPS (replaces 12 V SMPS + buck)	1	350–500
13	Backup cell	18650 Li-ion (fits the board's holder)	1	200–250
14	RTC module	DS3231	1	120–150
15	RS485 module	TTL ↔ RS485, auto direction	1	100–150
16	Relay module	1-channel 5 V opto-isolated	1	80–120
17	AC sense module	230 V opto-isolated, 8 channel (6 outgoing MCBs + contactor aux + spare)	1	400–500
17a	I/O expander	PCF8574 I2C module (reads the 6 MCB inputs)	1	80–120
18	Door limit switch		1	100–150
19	Antenna extension	SMA cable + panel-mount antenna	1	250–300
20	Terminal blocks	63 A for 16/25 sq mm × 4 (spec), control terminals	1 set	900–1,100

#	ITEM	SPEC / EXAMPLE	QTY	APPROX ₹
21	Wiring material	Cu wire, lugs, ferrules, DIN rail, duct, glands	1 set	1,500–2,000
22	Earthing	Earth bar, stud, washers	1 set	250–300
	Panel total (excluding poles, lights, cable, labour)			≈ 25,500–32,500

In parts me kanjoosi mat karna: spec me 5 saal ki comprehensive warranty hai. Contactor, MCB, RCCB, SMPS, earthing aur IP55 enclosure achhe brand ke lo. Yahan ₹500 bachaoge to baad me ₹5,000 site visit me lagenge.

6.1 Prototype ke liye extra (ek baar)

ITEM	QTY	APPROX ₹
USB-C data cable (flashing ke liye)	1	200
RS485 → USB adapter (meter ko PC se check karne ke liye)	1	250
Digital multimeter, clamp meter	1 each	2,500–4,000
500 V insulation tester (megger), earth tester, RCD tester	rent / electrician se	—
Test load: 6–10 LED lamps ya heater (bench test)	1 set	2,000–3,000

6.2 Har pole ka saamaan (spec ke hisaab se)

ITEM (PER POLE)	APPROX ₹
63 A terminal block (16/25 sq mm), 4 nos, DIN channel pe	400–600
6 A SP MCB C-curve, 2 nos	300–400
6 mm laminated sheet + mounting	100–200
Earth stud, washers, nuts, DIN channel	100–150
Fixing labour	150–300
Per pole	≈ 1,050–1,650

7. Working logic

7.1 Modes

MODE	KAISE CHALTA HAI
ASTRO (default)	Panel ki location se roz ka sunrise-sunset khud nikaalta hai. Sunset + 10 min ON, sunrise – 10 min OFF. Offset CMS se badal sakte ho.
SCHEDULE	Fixed time (jaise 18:30 ON, 06:00 OFF) aur hafte ke kaun se din.
MANUAL	CMS dashboard se turant ON/OFF. MQTT connection hamesha khula, command 1-2 second me lagta hai.
Special dates	8 tareekh tak alag time (Diwali, mela, election). ASTRO aur SCHEDULE ke upar chalta hai.

Selector switch AUTO pe na ho to controller sirf monitor aur report karta hai. Network chala jaye tab bhi lights panel ke andar ke timer se chalti rehti hain.

7.2 Har cycle me kya hota hai

STEP	KAAM
1	Boot: flash se settings, RTC se time. 4G se network time mile to RTC sync.
2	Har 5 second: meter padho (V, I, kW, PF per phase, total kW, Hz, kWh).
3	Selector AUTO hai? Nahi to step 6.
4	Aaj special date hai? Haan to uska time; nahi to mode (ASTRO / SCHEDULE / MANUAL) ke hisaab se ON ya OFF decide karo.
5	Relay se contactor ON/OFF.
6	Saare faults check karo (Section 7.3).
7	Koi fault naya aaya ya clear hua to turant CMS ko event bhejo.
8	Har 5 minute (CMS se badal sakte ho) poora data bhejo. Network na ho to flash me save, network aate hi bhejo.
—	CMS se command aaye to turant apply karo aur jawab bhejo.

7.3 Fault detection

FAULT	TELEMETRY KEY	HOW IT IS DETECTED	SEVERITY
Phase fail	f_phaseR/Y/B	Phase voltage below 180 V	Critical
Over voltage	f_overVolt	Any phase above 270 V	Major
Over current	f_overCurrent	Any phase above the set limit (default 25 A; setLimits for bigger panels)	Major
Lamp failure	f_lampR/Y/B	10 min after switching on, phase power below 95% of the learned baseline (LED drivers are constant-power, so power is steadier than current). Lamps failed $\approx$ (baseline kW – present kW) / one lamp's watts. The baseline is learned on the first night or via learnBaseline. With ~83 lamps per phase this catches about 4 or more failed lamps per phase; a single lamp is within meter accuracy.	Major
Outgoing MCB trip	f_mcb, mcbTripped (e.g. A-R, B-Y)	Phase voltage present and contactor on, but no 230 V after that MCB	Critical
Earth leakage (RCCB trip)	f_earthLeak	Contactor on and phases present, but all 6 MCB outputs dead	Critical
Contactor fault	f_contactor	Commanded ON, but no aux feedback or current after 30 s	Critical
Day burning	f_dayBurn	Commanded OFF, but current flowing for more than 1 min	Major
Door open	f_door	Door limit switch open	Major

FAULT	TELEMETRY KEY	HOW IT IS DETECTED	SEVERITY
Meter / supply fail	f_meter	No Modbus reply from the meter. The controller runs on its 18650 cell and still reports.	Critical
Network down	—	MQTT disconnected; data buffered in flash	Info

Limit: panel se ek-ek lamp ki kharabi nahi pakdi ja sakti. 83 lamp per phase pe ek phase me 4 ya zyada lamp band hon tabhi pata chalega. Kaun sa pole kharab hai, ye jaanne ke liye har lamp pe controller lagega, jo spec me nahi hai.

8. Firmware aur CMS

8.1 Firmware

File: `firmware/street_light_panel/street_light_panel.ino` (Arduino IDE). Sab 160 panels me **ek hi firmware** jaata hai. Har panel pehli baar online aate hi CMS me `CMS-<IMEI>` naam se khud register hota hai aur apna token flash me rakhta hai.

SETTING (FILE KE UPAR)	KYA DAALNA HAI
<code>BOARD_T_A7670</code> / <code>BOARD_T_SIM7600</code>	Kaun sa board (default T-A7670)
<code>METER_SELEC_MFM383A</code> / <code>METER_SDM630</code>	Kaun sa meter (default Selec)
<code>APN</code>	SIM ka APN (Jio <code>jionet</code> , Airtel <code>airtelgprs.com</code> , Vi <code>www</code> , BSNL <code>bsnlnet</code> , ya M2M SIM ka APN)
<code>MQTT_HOST</code> , <code>MQTT_PORT</code>	CMS server ka address, port 1883
<code>PROVISION_KEY</code> , <code>PROVISION_SECRET</code>	CMS me set kiye gaye provisioning key/secret
<code>DEFAULT_LAT</code> , <code>DEFAULT_LON</code>	Default location (har panel ki asli location baad me CMS se)

8.2 CMS se commands

METHOD	EXAMPLE PARAMS	EFFECT
<code>setLight</code>	<code>true</code>	Switch on now (sets MANUAL mode)
<code>setMode</code>	<code>"ASTRO"</code>	ASTRO / SCHEDULE / MANUAL
<code>setSchedule</code>	<code>{"on":"18:30","off":"06:00","days":127}</code>	days: bit0 = Sunday ... bit6 = Saturday (127 = every day)
<code>setAstro</code>	<code>{"onOffset":10,"offOffset":-10}</code>	Minutes after sunset / relative to sunrise
<code>setSpecial</code>	<code>{"slot":0,"date":"11-08","on":"17:30","off":"06:30"}</code>	Different times on one date (8 slots)
<code>learnBaseline</code>	<code>none</code>	Re-learn normal phase currents after lamps are replaced
<code>setInterval</code>	<code>300</code>	Seconds between normal data uploads (60–3600). Lower data cost with a longer interval.
<code>setLimits</code>	<code>{"iMax":25}</code>	Over-current limit per phase (25 A for 10 kW, 35 A for 15 kW, 50 A for 22.5 kW)
<code>setLampW</code>	<code>40</code>	Wattage of one lamp, for counting failed lamps
<code>setLocation</code>	<code>{"lat":28.61,"lon":77.21}</code>	Panel location for sunrise/sunset (set once per panel)
<code>getStatus</code>	<code>none</code>	Full live status in the reply

8.3 CMS software: ThingsBoard ya apna

Do raaste hain. Firmware dono ke saath same chalta hai; sirf `MQTT_HOST` aur provisioning key badalni hai.

	THINGSBOARD CE (FREE)	APNA CMS SOFTWARE (PR #2)
Kya hai	Ready-made open-source IoT platform	Is project ke liye bana Node.js server + web dashboard
Shuru karne me time	1–2 hafte	Phase 1 ready hai (repo me), baaki phases 6–8 hafte
Naam / logo	Nahi	Poora aapka
License	Free	Free, aapka product
Features (abhi)	Dashboard, alarms, history, map	Login + user roles, map, live data, history graph, CSV, alarms, event log, remote control, auto registration
Baaki	Scheduled reports paid version me	SMS/email alerts, group scheduling, PDF/Excel reports, Excel import, HTTPS guide

Apna software Docker se ek server pe chalta hai (`cms-server/`, `docker compose up -d`). Server kharcha 160 panel ke liye lagbhag ₹1.5–3.5 lakh, 5 saal. Tender me “hosting / data centre” clause check karna; kai govt tender data India me ya MeitY empanelled cloud pe maangte hain.

Recommendation: tender demo ke liye jo jaldi chal jaaye (dono ready hain), aur lambe samay ke liye apna software, kyunki license zero hai aur product aapka banta hai.

9. Panel banane ka tareeka (assembly)

1. **Enclosure layout:** upar incomer + SPD, beech me meter + busbar + contactor + RCCB, neeche outgoing MCBs + terminal blocks. Control compartment (controller, SMPS, modules) alag, power se door. Door pe selector switch aur indicator lamps.
2. **DIN rails aur wiring duct** lagao. Busbar SMC insulators pe, phase colour sleeve ke saath.
3. **Power wiring** (drawing 3): incomer → meter (CT R/Y/B pe, P1 supply side) → busbar → contactor → RCCB → 6 MCB → outgoing terminal blocks. Lugs crimp karo, sab bolt torque se tight.
4. **SPD:** incomer ke baad, chhote aur seedhe wire se earth bar tak (lambi earth wire SPD ko bekar karti hai).
5. **Control wiring** (drawing 4): 6 A MCB → selector → relay RL1 / direct → contactor A1, A2 → N. Aux NO → opto ch7.
6. **Opto sensing:** har outgoing MCB ke neeche wale terminal se opto ch1-6 (A-R, A-Y, A-B, B-R, B-Y, B-B order me), neutral common.
7. **Low-voltage wiring** (drawing 5): SMPS 5 V → LilyGO 5V/GND. RS485 module → meter A/B (twisted pair, end pe 120 Ω). DS3231 aur PCF8574 I2C pe. Relay module, door switch, selector contact. Pull-ups GPIO35/36 pe.
8. **Earthing:** panel body, door (braided strap), SPD, meter, earth bar → 2 earth pits. Har metal part earth se joda ho.
9. **Labels:** har wire pe ferrule number, har MCB pe naam (A-R, B-Y...), panel pe IMEI aur panel number.
10. **Cable entry:** neeche se glands ke through, IP55 bana rahe.

10. Testing procedure

Har test ke saamne result likho aur ✓ lagao. Har panel ki ek test sheet file karo (warranty ke liye kaam aayegi).

10.1 Bench test (pehla prototype, table pe)

#	TEST	KAISE KAREIN	EXPECTED RESULT	✓
B1	Firmware compile	Arduino IDE me Verify	Koi error nahi	☐
B2	Flash aur boot	USB se upload, Serial Monitor 115200	Boot message, modem start	☐
B3	Meter communication	Meter 230 V pe, RS485 joda	V, I, kW sahi; meter display se $\pm 1\%$	☐
B4	Word order (Selec)	Values galat/bahut bade aayein	WORD_SWAP badal ke dobara	☐
B5	4G + CMS	SIM daalo, power on	Dashboard pe CMS - <IMEI> khud aata hai	☐
B6	Location	setLocation bhejo	Sunrise/sunset sahi dikhta hai	☐
B7	Remote control	Dashboard switch ON/OFF	Relay 1-2 second me chale	☐
B8	Inputs	Door switch, selector, opto (230 V test lamp se)	CMS pe sahi state	☐
B9	Offline buffer	SIM nikaalo 15 min, phir lagao	Beech ka data CMS pe aata hai	☐
B10	Power fail	230 V hatao (18650 lagi ho)	Meter fail / supply off alarm CMS pe	☐

10.2 Panel test (har panel, factory me, bina load)

#	TEST	KAISE KAREIN	EXPECTED RESULT	✓
P1	Visual	Wiring drawing se milao, ferrules, torque	Sab sahi aur tight	☐
P2	Continuity	Multimeter se har power aur control wire	Har circuit sahi jagah	☐
P3	Insulation	500 V megger: phase-phase, phase-N, phase-earth (controller aur SMPS hata ke)	$\geq 1 \text{ M}\Omega$ (aam taur pe kaafi zyada)	☐
P4	Earth continuity	Body, door, SPD, earth bar ke beech	$\approx 0 \Omega$ ($< 0.5 \Omega$)	☐
P5	Phase sequence	Supply dene ke baad phase sequence meter	R-Y-B sahi, meter pe teeno V aayein	☐
P6	Selector MANUAL	Selector MAN	Contacto turant ON	☐
P7	Selector AUTO	AUTO + CMS se ON/OFF	Contacto command pe chale, aux feedback CMS pe	☐
P8	RCCB	Test button dabao; RCD tester se 300 mA	Trip ho; CMS pe earth leakage alarm	☐
P9	MCB sense	Har outgoing MCB ek-ek karke band	CMS pe sahi naam (A-R ... B-B)	☐
P10	Door	Door kholo	Door alarm	☐
P11	Soak test	24 ghante chalao	Koi reboot nahi, data har 5 min	☐

10.3 Load test (prototype ya sample panel)

#	TEST	KAISE KAREIN	EXPECTED RESULT	✓
L1	Load ON	Asli feeder ya test load se ON	kW, PF sahi; teeno phase balance	☐

#	TEST	KAISE KAREIN	EXPECTED RESULT	✓
L2	Baseline	10 min baad (ya learnBaseline)	Base power save	☐
L3	Lamp failure	Ek phase pe 4-5 lamp band	Lamp failure alarm, approx lamp count	☐
L4	Over current	setLimits me iMax kam karke	Over current alarm	☐
L5	Day burning	AUTO me OFF time pe MANUAL kar do	1 min baad day burning alarm	☐
L6	Contactor fault	Coil ka wire nikaal ke ON command	30 s baad contactor fault	☐
L7	ASTRO	1-2 hafte chalao	Roz sahi time pe ON/OFF, event log me record	☐

10.4 Site commissioning (har panel)

#	KAAM	EXPECTED RESULT	✓
S1	Plinth / foundation, panel level aur fix	Mazboot, paani na ghuse	☐
S2	2 earth pits, earth resistance test	Kam se kam (aam taur pe < 5 Ω, ya tender ke hisaab se)	☐
S3	Incoming aur feeder cable termination, glands	Tight, IP55 bana rahe	☐
S4	Feeder insulation test (lamps hata ke ya MCB off)	≥ 1 MΩ	☐
S5	Lamps R-Y-B baari baari jude hain	Teeno phase ka kW lagbhag barabar	☐
S6	Power on, CMS pe panel online	CMS - <IMEI> online, signal (rssi) theek	☐
S7	Location aur naam set	Map pe sahi jagah, ward/road naam	☐
S8	Remote ON/OFF, MANUAL selector	Dono kaam karein	☐
S9	Voltage drop	Feeder ke last pole pe voltage	Supply se ~5% se kam gira
S10	Handover	Test sheet, IMEI, photos file	Record complete

11. Bid estimate (160 panels + CMS, 5 saal)

ITEM	PER PANEL ₹	160 PANELS ₹
Panel material (10 kW tender build, ~10% bulk discount)	23,000–29,500	36.8–47.2 lakh
Assembly, wiring, testing per panel	2,500–3,500	4.0–5.6 lakh
Transport	500–1,000	0.8–1.6 lakh
Site installation and commissioning (plinth, 2 earth pits, terminations)	8,000–12,000	12.8–19.2 lakh
SIM data, 5 years (M2M plan)	2,400–6,000	3.8–9.6 lakh
CMS server 5 years + dashboard setup	—	2.5–4.0 lakh
Spares and warranty visits reserve (~10% of panel material)	2,300–2,950	3.7–4.7 lakh
Prototype and development (one-time)	—	1.0–2.0 lakh
Total before profit and GST		≈ ₹65–94 lakh

Iske upar apna margin aur 18% GST. 60 W ya 90 W lamp ho to rating table ka extra cost jodo (160 panel pe ₹4.8–9.6 lakh). Feeder cable aur poles isme nahi hain. Spec ka per-pole saamaan agar sab 40,000 pole pe lagna hai to ≈ ₹4.2–6.6 crore alag.

12. Safety

415 V jaan le sakta hai. Power wiring aur testing licensed electrician se karwao. Panel body aur door earth karo. Panel ke andar kaam se pehle incomer OFF karke lock karo (lock-out). Megger test se pehle controller, SMPS aur opto module ke wire hata do.

13. Abhi kya tay karna baaki hai

1. Har pole pe lamp kitne watt ka hai (panel rating isi se tay hogi).
2. Naye pole kitne hain, aur window box ka saamaan sab 40,000 pole pe lagna hai ya sirf naye pe.
3. CMS: ThingsBoard ya apna software, aur server kahan host hoga (tender ka hosting clause).
4. Firmware ka pehla compile aur bench test (Section 10.1); Selec meter ka word order confirm.

Appendix A. Firmware code

```
/*
Smart Feeder Control Panel (Street Light) -> CMS dashboard
-----
Board : LilyGO T-A7670E (ESP32 + 4G LTE Cat-1, sasta) [ya T-SIM7600E]
Meter : Selec MFM383A-C (3-phase, RS485, slave id 1, 9600 8N1) [ya Eastron SDM630]
RTC : DS3231 (I2C)
CMS : ThingsBoard (MQTT, device access token)

Libraries (Arduino Library Manager se install karo):
TinyGSM, PubSubClient, ModbusMaster, RTCLib, ArduinoJson (v7)
(T-A7670 ke liye agar TinyGSM me A7670 na mile to LilyGO-T-A76XX repo
ke "lib" folder wali TinyGSM use karo)
Board package: "esp32 by Espressif", board = "ESP32 Dev Module",
Partition scheme = "Default 4MB with spiffs" (LittleFS ke liye)

NOTE: Pin numbers LilyGO T-A7670 / T-SIM7600 ke hisaab se hain. Apne board ka
pinout zaroor check karo. GPIO 34-39 input-only hain, inpe
10k external pull-up lagana hai.
*/

// ---- Hardware choice: ek board aur ek meter chuno ----
#define BOARD_T_A7670 // sasta (tender default). Mehenga option: BOARD_T_SIM7600
#define METER_SELEC_MFM383A // sasta (tender default). Mehenga option: METER_SDM630

#if defined(BOARD_T_A7670)
#define TINY_GSM_MODEM_A7670
#else
#define TINY_GSM_MODEM_SIM7600
#endif
#define TINY_GSM_RX_BUFFER 1024
#include <TinyGsmClient.h>
#include <PubSubClient.h>
#include <ModbusMaster.h>
#include <RTCLib.h>
#include <ArduinoJson.h>
#include <Preferences.h>
#include <LittleFS.h>
#include <math.h>

// ===== SETTINGS (apne hisaab se badlo) =====
const char APN[] = "airtelgprs.com"; // Jio: "jionet", Vi: "www", BSNL: "bsnlnet"
const char MQTT_HOST[] = "cms.example.com"; // aapka ThingsBoard server
const uint16_t MQTT_PORT = 1883;
// 160 panels me SAME firmware jaata hai. Har panel pehli baar online aate hi
// ThingsBoard "device provisioning" se apna token khud le leta hai (naam = CMS-<IMEI>).
// Location CMS se setLocation command se set hoti hai (ya neeche default).
const char PROVISION_KEY[] = "cms-provision-key"; // ThingsBoard device profile me set karo
const char PROVISION_SECRET[] = "cms-provision-secret";
const float DEFAULT_LAT = 28.61, DEFAULT_LON = 77.21; // jab tak setLocation na aaye (example: Delhi)
const float TZ_HOURS = 5.5; // IST

// Fault limits
const float V_LOW = 180, V_HIGH = 270; // volts (phase-neutral)
// LED driver constant-power hote hain, isliye lamp failure POWER (kW) se check hota hai,
// current se nahi. ~83 lamp/phase pe 5% drop = ~4 lamp band. Isse chhota farq meter
// accuracy (class 1) me chhup jaata hai.
const float LAMP_DROP = 0.95;
const uint32_t SETTLE_MS = 10UL * 60 * 1000; // ON hone ke 10 min baad hi lamp check

// ===== PINS (LilyGO T-A7670 / T-SIM7600) =====
#define MODEM_TX 27
#define MODEM_RX 26
#define MODEM_PWRKEY 4
#if defined(BOARD_T_A7670)
#define MODEM_POWERON 12 // board ka modem power enable
#else
#define MODEM_FLIGHT 25
#endif
#define RS485_RX 18 // TTL-RS485 module (auto direction) ka TXD
#define RS485_TX 19 // TTL-RS485 module ka RXD
#define I2C_SDA 21 // DS3231
#define I2C_SCL 22
#define PIN_RELAY 23 // relay -> contactor coil (AUTO path)
#define PIN_DOOR 35 // door limit switch, LOW = door band
#define PIN_AUX 36 // contactor aux NO via 230V opto, LOW = contactor ON
// 6 outgoing MCB (Feeder A: R,Y,B Feeder B: R,Y,B) ke baad 230V opto -> PCF8574
// I2C expander (RTC wali I2C line pe). Bit 0-5 = A-R, A-Y, A-B, B-R, B-Y, B-B.
// Bit LOW = MCB ke baad supply hai.
#define PCF8574_ADDR 0x20
#define PIN_SEL_AUTO 15 // selector switch AUTO position, LOW = AUTO
#define RELAY_ON_LEVEL HIGH

// ===== OBJECTS =====
HardwareSerial SerialAT(1);
TinyGsm modem(SerialAT);
TinyGsmClient gsmClient(modem);
PubSubClient mqtt(gsmClient);
ModbusMaster meterBus;
RTC_DS3231 rtc;
Preferences prefs;

// ===== DATA =====
struct Meter {
float v[3], i[3], p[3], pf[3];
float kwTotal, freq, kwh;
bool ok;
};
```

```

} m;

struct Special { uint16_t mddd; int16_t onMin, offMin; }; // mddd=0 means empty

enum Mode : uint8_t { MODE_ASTRO = 0, MODE_SCHEDULE = 1, MODE_MANUAL = 2 };

struct Config {
    uint8_t mode;
    int16_t onOffset, offOffset; // ASTRO: sunset+onOffset ON, sunrise+offOffset OFF (minutes)
    int16_t onMin, offMin; // SCHEDULE: minutes from midnight
    uint8_t dayMask; // bit0=Sun ... bit6=Sat
    bool manualState;
    float basePw[3]; // normal power per phase (kW), auto-learn
    float lampW; // ek lamp ki wattage (W), e.g. 40
    float lat, lon; // panel location (ASTRO ke liye)
    float iMax; // over-current limit per phase (A): 10kW=25, 15kW=35, 22kW=50
    char token[40]; // ThingsBoard access token (provisioning se aata hai)
    uint16_t publishSec; // normal data kitne second me bheje (data/SIM kharcha kam)
    Special specials[8]; // date-wise override (festival etc.)
} cfg;

enum Fault : uint32_t {
    F_PHASE_R = 1UL << 0, F_PHASE_Y = 1UL << 1, F_PHASE_B = 1UL << 2,
    F_OVERVOLT = 1UL << 3, F_OVERCURRENT = 1UL << 4,
    F_LAMP_R = 1UL << 5, F_LAMP_Y = 1UL << 6, F_LAMP_B = 1UL << 7,
    F_CONTACTOR = 1UL << 8, F_DAY_BURN = 1UL << 9,
    F_MCB = 1UL << 10, F_EARTH_LEAK = 1UL << 11,
    F_DOOR = 1UL << 12, F_METER = 1UL << 13
};
const char* FAULT_KEYS[] = {
    "f_phaseR", "f_phaseY", "f_phaseB", "f_overVolt", "f_overCurrent",
    "f_lampR", "f_lampY", "f_lampB", "f_contactor", "f_dayBurn",
    "f_mcb", "f_earthLeak", "f_door", "f_meter"
};
const int FAULT_COUNT = sizeof(FAULT_KEYS) / sizeof(FAULT_KEYS[0]);

uint32_t faults = 0, lastFaults = 0;
bool lightOn = false;
uint32_t onSinceMs = 0, dayBurnSinceMs = 0;
int sunriseMin = 360, sunsetMin = 1080;
int failedLamps[3] = {0, 0, 0};
String mcbTripped = ""; // jaise "A-R,B-Y"
uint32_t tRead = 0, tPublish = 0, tConnect = 0;

// ===== TIME HELPERS =====
DateTime localNow() { return rtc.now() + TimeSpan((int32_t)(TZ_HOURS * 3600)); }
uint64_t epochMs() { return (uint64_t)rtc.now().unixtime() * 1000ULL; }

int dayOfYear(const DateTime& d) {
    static const int cum[] = {0, 31, 59, 90, 120, 151, 181, 212, 243, 273, 304, 334};
    int n = cum[d.month() - 1] + d.day();
    bool leap = (d.year() % 4 == 0 && d.year() % 100 != 0) || d.year() % 400 == 0;
    return (leap && d.month() > 2) ? n + 1 : n;
}

// Sunrise/sunset (NOAA "Almanac" method). Returns local minutes from midnight, -1 if none.
int sunEvent(int doy, bool rising) {
    const float RAD = M_PI / 180.0, ZENITH = 90.833;
    float lngHour = cfg.lon / 15.0;
    float t = doy + ((rising ? 6 : 18) - lngHour) / 24.0;
    float M = 0.9856 * t - 3.289;
    float L = fmod(M + 1.916 * sin(M * RAD) + 0.020 * sin(2 * M * RAD) + 282.634 + 360, 360);
    float RA = fmod(atan(0.91764 * tan(L * RAD)) / RAD + 360, 360);
    RA += floor(L / 90) * 90 - floor(RA / 90) * 90;
    RA /= 15;
    float sinDec = 0.39782 * sin(L * RAD);
    float cosDec = cos(asin(sinDec));
    float cosH = (cos(ZENITH * RAD) - sinDec * sin(cfg.lat * RAD)) / (cosDec * cos(cfg.lat * RAD));
    if (cosH > 1 || cosH < -1) return -1;
    float H = (rising ? 360 - acos(cosH) / RAD : acos(cosH) / RAD) / 15;
    float T = H + RA - 0.06571 * t - 6.622;
    float local = fmod(T - lngHour + TZ_HOURS + 48, 24);
    return (int)(local * 60);
}

// ON window jo midnight cross kar sakti hai (e.g. 18:30 -> 06:00)
bool inWindow(int now, int on, int off) {
    return (on <= off) ? (now >= on && now < off) : (now >= on || now < off);
}

int parseHHMM(const char* s) { int h = 0, mi = 0; sscanf(s, "%d:%d", &h, &mi); return h * 60 + mi; }

// ===== CONFIG =====
void loadConfig() {
    prefs.begin("panel", false);
    if (prefs.getBytes("cfg", &cfg, sizeof(cfg)) != sizeof(cfg)) {
        memset(&cfg, 0, sizeof(cfg));
        cfg.mode = MODE_ASTRO;
        cfg.onOffset = 10; // sunset ke 10 min baad ON
        cfg.offOffset = -10; // sunrise se 10 min pehle OFF
        cfg.onMin = 18 * 60 + 30;
        cfg.offMin = 6 * 60;
        cfg.dayMask = 0x7F;
        cfg.lampW = 40;
        cfg.lat = DEFAULT_LAT; cfg.lon = DEFAULT_LON;
        cfg.iMax = 25;
        cfg.publishSec = 300; // 5 min. Faults/events phir bhi turant jaate hain
    }
}

void saveConfig() { prefs.putBytes("cfg", &cfg, sizeof(cfg)); }

```

```

// ===== METER (Modbus RS485) =====
// Register map (input registers, function 04, float = 2 registers).
// Selec ka map aggssoft.com ki list se hai; word order meter ke manual ya
// "Modbus Poll" software se ek baar check kar lo (galat ho to WORD_SWAP badlo).
#if defined(METER_SDM630)
    const uint16_t R_V[3] = {0, 2, 4}, R_I[3] = {6, 8, 10}, R_P[3] = {12, 14, 16}, R_PF[3] = {30, 32, 34};
    const uint16_t R_KW = 52, R_FREQ = 70, R_KWH = 72;
    const float P_DIV = 1000.0; // SDM630 power W me deta hai
    const bool WORD_SWAP = false; // high word pehle
#else // Selec MFM383A-C
    const uint16_t R_V[3] = {0, 2, 4}, R_I[3] = {16, 18, 20}, R_P[3] = {24, 26, 28}, R_PF[3] = {48, 50, 52};
    const uint16_t R_KW = 42, R_FREQ = 56, R_KWH = 58;
    const float P_DIV = 1.0; // kW me
    const bool WORD_SWAP = true; // low word pehle (verify karo)
#endif

bool readFloat(uint16_t reg, float& out) {
    if (meterBus.readInputRegisters(reg, 2) != meterBus.ku8MBSuccess) return false;
    uint16_t a = meterBus.getResponseBuffer(0), b = meterBus.getResponseBuffer(1);
    uint32_t raw = WORD_SWAP ? ((uint32_t)b << 16) | a : ((uint32_t)a << 16) | b;
    memcpy(&out, &raw, 4);
    return true;
}

void readMeter() {
    bool ok = true;
    for (int k = 0; k < 3; k++) {
        ok &= readFloat(R_V[k], m.v[k]);
        ok &= readFloat(R_I[k], m.i[k]);
        ok &= readFloat(R_P[k], m.p[k]); m.p[k] /= P_DIV;
        ok &= readFloat(R_PF[k], m.pf[k]);
    }
    ok &= readFloat(R_KW, m.kwTotal); m.kwTotal /= P_DIV;
    ok &= readFloat(R_FREQ, m.freq);
    ok &= readFloat(R_KWH, m.kwh);
    m.ok = ok;
}

// ===== LIGHT DECISION =====
bool selectorAuto() { return digitalRead(PIN_SEL_AUTO) == LOW; }

void setLight(bool on) {
    if (on && !lightOn) onSinceMs = millis();
    lightOn = on;
    digitalWrite(PIN_RELAY, on ? RELAY_ON_LEVEL : !RELAY_ON_LEVEL);
}

void decideLight() {
    DateTime d = localNow();
    int nowMin = d.hour() * 60 + d.minute();
    int doy = dayOfYear(d);
    sunriseMin = sunEvent(doy, true);
    sunsetMin = sunEvent(doy, false);

    bool want = false;
    uint16_t mmdd = d.month() * 100 + d.day();
    const Special* sp = nullptr;
    for (auto& s : cfg.specials) if (s.mmdd == mmdd) sp = &s;

    if (cfg.mode == MODE_MANUAL) {
        want = cfg.manualState;
    } else if (sp) {
        want = inWindow(nowMin, sp->onMin, sp->offMin);
    } else if (cfg.mode == MODE_ASTRO) {
        want = inWindow(nowMin, sunsetMin + cfg.onOffset, sunriseMin + cfg.offOffset);
    } else { // SCHEDULE
        // subah ka hissa (midnight ke baad) pichle din ki schedule maana jaata hai
        int dow = d.dayOfTheWeek();
        if (cfg.onMin > cfg.offMin && nowMin < cfg.offMin) dow = (dow + 6) % 7;
        want = (cfg.dayMask & (1 << dow)) && inWindow(nowMin, cfg.onMin, cfg.offMin);
    }
    setLight(want);
}

// ===== FAULT DETECTION =====
uint8_t readExpander() {
    if (Wire.requestFrom(PCF8574_ADDR, 1) != 1) return 0x00; // expander na mile to "sab theek" maano
    return Wire.read();
}

void checkFaults() {
    uint32_t f = 0;
    const uint32_t phaseBits[3] = {F_PHASE_R, F_PHASE_Y, F_PHASE_B};
    const uint32_t lampBits[3] = {F_LAMP_R, F_LAMP_Y, F_LAMP_B};
    static const char* MCB_NAMES[6] = {"A-R", "A-Y", "A-B", "B-R", "B-Y", "B-B"};

    if (digitalRead(PIN_DOOR) == HIGH) f |= F_DOOR;
    if (!m.ok) { f |= F_METER; faults = f; return; }

    bool contactorOn = digitalRead(PIN_AUX) == LOW;
    bool settled = lightOn && (millis() - onSinceMs > SETTLE_MS);
    float totalI = 0;

    for (int k = 0; k < 3; k++) {
        totalI += m.i[k];
        if (m.v[k] < V_LOW) f |= phaseBits[k];
        if (m.v[k] > V_HIGH) f |= F_OVERVOLT;
        if (m.i[k] > cfg.iMax) f |= F_OVERCURRENT;
        bool phaseOk = m.v[k] >= V_LOW;
    }

```

```

// Lamp failure: phase ki power normal (base) se 5% se zyada kam
failedLamps[k] = 0;
if (settled && phase0k && cfg.basePw[k] > 0.1 && m.p[k] < cfg.basePw[k] * LAMP_DROP) {
    f |= lampBits[k];
    failedLamps[k] = (int)roundf((cfg.basePw[k] - m.p[k]) * 1000.0 / cfg.lampW);
}
}

// Outgoing MCB / RCCB trip: phase hai, contactor ON hai, par MCB ke baad supply nahi
mcbTripped = "";
if (contactorOn) {
    uint8_t in = readExpander();
    int dead = 0, live = 0;
    for (int j = 0; j < 6; j++) {
        if (m.v[j] % 3 < V_LOW) continue; // phase hi nahi hai, wo phase fault hai
        live++;
        if (in & (1 << j)) { // HIGH = supply nahi
            dead++;
            if (mcbTripped.length()) mcbTripped += ",";
            mcbTripped += MCB_NAMES[j];
        }
    }
    if (live > 0 && dead == live) f |= F_EARTH_LEAK; // sab feeder band = RCCB trip
    else if (dead > 0) f |= F_MCB;
}

// Pehli baar: base power khud seekh lo
if (settled && cfg.basePw[0] == 0 && cfg.basePw[1] == 0 && cfg.basePw[2] == 0 && totalI > 1.0) {
    for (int k = 0; k < 3; k++) cfg.basePw[k] = m.p[k];
    saveConfig();
}

// Contactor fault: command ON tha par contactor/current nahi aaya
if (selectorAuto() && lightOn && millis() - onSinceMs > 30000 && (!contactorOn || totalI < 0.3))
    f |= F_CONTACTOR;

// Day burning: command OFF hai par lights jal rahi hain (1 min se zyada)
if (selectorAuto() && !lightOn && totalI > 0.5) {
    if (!dayBurnSinceMs) dayBurnSinceMs = millis();
    if (millis() - dayBurnSinceMs > 60000) f |= F_DAY_BURN;
} else dayBurnSinceMs = 0;

faults = f;
}

// ===== TELEMETRY + OFFLINE LOG =====
void buildTelemetry(JsonObject v) {
    const char* ph[3] = {"R", "Y", "B"};
    for (int k = 0; k < 3; k++) {
        v[String("v") + ph[k]] = m.v[k];
        v[String("i") + ph[k]] = m.i[k];
        v[String("kw") + ph[k]] = m.p[k];
        v[String("pf") + ph[k]] = m.pf[k];
        v[String("lampsFailed") + ph[k]] = failedLamps[k];
    }
    v["kwTotal"] = m.kwTotal; v["freq"] = m.freq; v["kwh"] = m.kwh;
    v["light"] = lightOn;
    v["mode"] = cfg.mode == MODE_ASTR0 ? "ASTRO" : cfg.mode == MODE_SCHEDULE ? "SCHEDULE" : "MANUAL";
    v["selector"] = selectorAuto() ? "AUTO" : "LOCAL";
    v["sunrise"] = sunriseMin; v["sunset"] = sunsetMin;
    v["rssi"] = modem.getSignalQuality();
    for (int b = 0; b < FAULT_COUNT; b++) v[FAULT_KEYS[b]] = (bool)(faults & (1UL << b));
    v["mcbTripped"] = mcbTripped;
}

void logOffline(const String& line) {
    File fl = LittleFS.open("/queue.jsonl", FILE_APPEND);
    if (fl && fl.size() < 900000) fl.println(line); // ~1MB tak store
    if (fl) fl.close();
}

void flushOffline() {
    if (!LittleFS.exists("/queue.jsonl")) return;
    File fl = LittleFS.open("/queue.jsonl", FILE_READ);
    while (fl.available() && mqtt.connected()) {
        String line = fl.readStringUntil('\n');
        line.trim();
        if (line.length()) mqtt.publish("v1/devices/me/telemetry", line.c_str());
        mqtt.loop();
    }
    fl.close();
    LittleFS.remove("/queue.jsonl");
}

void publishTelemetry() {
    JsonDocument doc;
    doc["ts"] = epochMs();
    buildTelemetry(doc["values"].to<JsonObject>());
    String out; serializeJson(doc, out);
    if (mqtt.connected()) mqtt.publish("v1/devices/me/telemetry", out.c_str());
    else logOffline(out);
}

// Event log: har fault RAISED / CLEARED turant bhejo
void publishEvents() {
    uint32_t changed = faults ^ lastFaults;
    if (!changed) return;
    for (int b = 0; b < FAULT_COUNT; b++) {
        if (!(changed & (1UL << b))) continue;
        JsonDocument doc;
        doc["ts"] = epochMs();
    }
}

```

```

        doc["values"]["event"] = String(FAULT_KEYS[b]) + ((faults & (1UL << b)) ? " RAISED" : " CLEARED");
        String out; serializeJson(doc, out);
        if (mqtt.connected()) mqtt.publish("v1/devices/me/telemetry", out.c_str());
        else logOffline(out);
    }
    lastFaults = faults;
    publishTelemetry(); // alarm ke saath latest values bhi
}

// ===== COMMANDS FROM CMS (RPC) =====
bool provisioned = false;

void onMqtt(char* topic, byte* payload, unsigned int len) {
    String t(topic);
    if (t == "/provision/response") { // {"status":"SUCCESS","credentialsValue":"..."}
        JsonDocument r;
        if (!deserializeJson(r, payload, len) && String(r["status"] | "") == "SUCCESS") {
            strcpy(cfg.token, r["credentialsValue"] | "", sizeof(cfg.token));
            saveConfig();
            provisioned = true;
        }
        return;
    }
    String reqId = t.substring(t.lastIndexOf('/') + 1);
    JsonDocument req;
    if (deserializeJson(req, payload, len)) return;
    String method = req["method"] | "";
    JsonVariant p = req["params"];
    JsonDocument res; res["ok"] = true;

    if (method == "setLight") { // {"method":"setLight","params":true}
        cfg.mode = MODE_MANUAL; cfg.manualState = p.as<bool>();
    } else if (method == "setMode") { // "ASTRO" | "SCHEDULE" | "MANUAL"
        String md = p.as<String>();
        cfg.mode = md == "SCHEDULE" ? MODE_SCHEDULE : md == "MANUAL" ? MODE_MANUAL : MODE_ASTRO;
    } else if (method == "setSchedule") { // {"on":"18:30","off":"06:00","days":127}
        cfg.onMin = parseHMM(p["on"] | "18:30");
        cfg.offMin = parseHMM(p["off"] | "06:00");
        cfg.dayMask = p["days"] | 0x7F;
    } else if (method == "setAstro") { // {"onOffset":10,"offOffset":-10}
        cfg.onOffset = p["onOffset"] | 0; cfg.offOffset = p["offOffset"] | 0;
    } else if (method == "setSpecial") { // {"slot":0,"date":"10-31","on":"17:30","off":"06:30"}
        int slot = p["slot"] | 0;
        int mo = 0, dy = 0; sscanf(p["date"] | "00-00", "%d-%d", &mo, &dy);
        if (slot >= 0 && slot < 8) cfg.specials[slot] = {(uint16_t)(mo * 100 + dy),
            (int16_t)parseHMM(p["on"] | "00:00"), (int16_t)parseHMM(p["off"] | "00:00")};
    } else if (method == "learnBaseline") { // naye lamps lagne ke baad
        for (int k = 0; k < 3; k++) cfg.basePw[k] = m.p[k];
    } else if (method == "setInterval") { // seconds, 60..3600
        cfg.publishSec = constrain(p.as<int>(), 60, 3600);
    } else if (method == "setLimits") { // {"iMax":25}
        cfg.iMax = p["iMax"] | cfg.iMax;
    } else if (method == "setLampW") { // 40
        cfg.lampW = p.as<float>();
    } else if (method == "setLocation") { // {"lat":28.61,"lon":77.21}
        cfg.lat = p["lat"] | cfg.lat; cfg.lon = p["lon"] | cfg.lon;
    } else if (method == "getStatus") {
        buildTelemetry(res["status"].to<JsonObject>());
    } else {
        res["ok"] = false; res["error"] = "unknown method";
    }
    saveConfig();
    decideLight(); // command turant apply, koi delay nahi

    String out; serializeJson(res, out);
    mqtt.publish(("v1/devices/me/rpc/response/" + reqId).c_str(), out.c_str());
    publishTelemetry();
}

// ===== NETWORK =====
void syncTimeFromNetwork() {
    int y, mo, d, h, mi, s; float tz;
    if (modem.getNetworkTime(&y, &mo, &d, &h, &mi, &s, &tz) && y >= 2025) {
        DateTime localT(y, mo, d, h, mi, s);
        rtc.adjust(localT - TimeSpan((int32_t)(tz * 3600))); // RTC me UTC rakhte hain
    }
}

void ensureConnected() {
    if (mqtt.connected()) return;
    if (millis() - tConnect < 30000 && tConnect) return; // har 30 s retry
    tConnect = millis();
    if (!modem.isNetworkConnected() && !modem.waitForNetwork(20000)) return;
    if (!modem.isGprsConnected() && !modem.gprsConnect(APN)) return;
    syncTimeFromNetwork();
    mqtt.setServer(MQTT_HOST, MQTT_PORT);
    mqtt.setCallback(onMqtt);
    mqtt.setKeepAlive(30);
    mqtt.setBufferSize(1024);
    String imei = modem.getIMEI();
    String clientId = "CMS-" + imei; // har panel ka alag client id

    // Pehli baar: ThingsBoard se apna token lo
    if (cfg.token[0] == 0) {
        if (!mqtt.connect(clientId.c_str(), "provision", nullptr)) return;
        mqtt.subscribe("/provision/response");
        JsonDocument req;
        req["deviceName"] = clientId;
        req["provisionDeviceKey"] = PROVISION_KEY;
        req["provisionDeviceSecret"] = PROVISION_SECRET;
        String out; serializeJson(req, out);
    }
}

```

```

    provisioned = false;
    mqtt.publish("/provision/request", out.c_str());
    for (uint32_t t0 = millis(); !provisioned && millis() - t0 < 15000;) mqtt.loop();
    mqtt.disconnect();
    if (!provisioned) return;           // 30 s baad phir try
}

if (mqtt.connect(clientId.c_str(), cfg.token, nullptr)) {
    mqtt.subscribe("v1/devices/me/rpc/request/+", 1);
    flushOffline();
    publishTelemetry();
} else if (mqtt.state() == MQTT_CONNECT_UNAUTHORIZED) {
    cfg.token[0] = 0; // CMS pe device delete hua: agli baar dobara provision
    saveConfig();
}

// ===== SETUP / LOOP =====
void setup() {
    Serial.begin(115200);
    pinMode(PIN_RELAY, OUTPUT); digitalWrite(PIN_RELAY, !RELAY_ON_LEVEL);
    pinMode(PIN_DOOR, INPUT); pinMode(PIN_AUX, INPUT); // external pull-up
    pinMode(PIN_SEL_AUTO, INPUT_PULLUP);

    Wire.begin(I2C_SDA, I2C_SCL);
    Wire.beginTransmission(PCF8574_ADDR); Wire.write(0xFF); Wire.endTransmission(); // sab pins input
    rtc.begin();
    LittleFS.begin(true);
    loadConfig();

    Serial2.begin(9600, SERIAL_8N1, RS485_RX, RS485_TX);
    meterBus.begin(1, Serial2);

    // 4G modem power on
    #if defined(BOARD_T_A7670)
        pinMode(MODEM_POWERON, OUTPUT); digitalWrite(MODEM_POWERON, HIGH);
        pinMode(MODEM_PWRKEY, OUTPUT);
        digitalWrite(MODEM_PWRKEY, LOW); delay(100); digitalWrite(MODEM_PWRKEY, HIGH); delay(100); digitalWrite(MODEM_PWRKEY, LOW);
    #else
        pinMode(MODEM_PWRKEY, OUTPUT);
        digitalWrite(MODEM_PWRKEY, HIGH); delay(300); digitalWrite(MODEM_PWRKEY, LOW);
        pinMode(MODEM_FLIGHT, OUTPUT); digitalWrite(MODEM_FLIGHT, HIGH);
    #endif
    SerialAT.begin(115200, SERIAL_8N1, MODEM_RX, MODEM_TX);
    delay(3000);
    modem.restart();

    readMeter(); decideLight(); checkFaults(); // network na ho tab bhi lights chalti rahein
}

void loop() {
    mqtt.loop();
    if (millis() - tRead > 5000) { // commands turant receive
        tRead = millis(); // har 5 s
        readMeter(); decideLight(); checkFaults(); publishEvents();
    }
    if (millis() - tPublish > cfg.publishSec * 1000UL) { // default har 5 min data log
        tPublish = millis();
        publishTelemetry();
    }
    ensureConnected();
}

```